INCH-POUND

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DETAIL SPECIFICATION

PACKING MATERIAL, GRAPHITIC OR CARBON BRAIDED YARN

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for staple or continuous filament graphitic or carbon yarn packing. Packing is for steam valve service on naval ships.

1.2 Classification. The material shall be of the following types, classes, and grades, as specified (see 3.1 and 6.2).

1.2.1 Types.

Type I – Continuous filament graphitic or carbon yarn
Type II – Staple graphitic or carbon yarn

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or e-mailed to commandstandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.2.2 Classes.

Class 1 – For use where detrimental material and lubricant content of the packing need not be controlled beyond normal manufacturing limits.
Class 2 – For use where detrimental material content must be controlled to limits specified herein.
Class 3 – For use where detrimental material content need not be controlled beyond normal manufacturing limits, and stuffing box temperatures do not exceed 260 degrees Celsius (°C) (500 degrees Fahrenheit (°F)).

1.2.3 Grades.

Grade I – Treated with corrosion inhibitor
Grade N – No corrosion inhibitor

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in Sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in Sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-45208	--	Inspection System Requirements.
MIL-L-19140	–	Lumber and Plywood, Fire-Retardant Treated.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2073-1	–	Department of Defense Standard Practices for Military Packaging.

(Copies of these documents are available online at http://quicksearch.dla.mil or https://assist.dla. mil or from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND

STR-ISO 9000 Supplement A –	Supplement to ISO 9000 (ANSI/ISO/ASQC Q9000) Quality Systems for Critical Procurements and Complex Procurements
STR-ISO 9000 Supplement B –	Supplement to ISO 9000 (ANSI/ISO/ASQC Q9000) Quality Systems for Critical Procurements and Not Complex Procurements
STR-ISO 9000 Supplement C –	Supplement to ISO 9000 (ANSI/ISO/ASQC Q9000) Quality Systems for Basic Procurements and Complex Procurements

(Copies of other Government publications should be obtained from the contracting agency or as directed by the contracting officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ISO/ASQ Q9001	–	Quality Management Systems – Requirements
ANSI/ISO/ASQ Q9002	–	Quality Systems – Model for Quality Assurance in Production Installation, and Service
ANSI/ISO/ASQ Q9003	–	Quality Systems – Model for Quality Assurance in Final Inspection and Test
ANSI/ASQ Z1.4	–	Sampling Procedures and Tables for Inspection by Attributes

(Copies of these documents are available from www.ansi.org or the American National Standards Institute, 25 West 43rd Street, New York, NY 10036.)

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ASTM INTERNATIONAL (ASTM)

C135 – Standard Test Method for True Specific Gravity of Refractory Materials by Water Immersion
C561 – Standard Test Method for Ash in Graphite Sample
C562 – Standard Test Method for Moisture in a Graphite Sample
C816 – Test Method for Sulfur in Graphite by Combustion - Iodometric Titration Method
C1502 -- Standard Test Method for Determination of Total Chloride and Fluorine in Uranium Dioxide and Gadolinium Oxide
D129 – Test Method for Sulfur in Petroleum Products (General High Pressure Decomposition Device Method)
D512 – Standard Test Methods for Chloride Ion in Water. (DoD adopted)
D516 -- Standard Test Method for Sulfate Ion in Water
D808 -- Standard Test Method for Chlorine in New and Used Petroleum Products (High Pressure Decomposition Device Method)
D1179 – Standard Test Methods for Fluoride Ion in Water
D1234 – Standard Method of Sampling and Testing Staple Length of Grease Wool
D3684 – Standard Test Method for Total Mercury in Coal by the Oxygen Bomb Combustion/Atomic Absorption Method
D3761 – Standard Test Method for Total Fluorine in Coal by the Oxygen Bomb Combustion/Ion Selective Electrode Method
D3951 – Standard Practice for Commercial Packaging. (DoD adopted)
D4239 – Standard Test Method for Sulfur in the Analysis Sample of Coal and Coke Using High Temperature Tube Furnace Combustion Methods
D4327 -- Standard Test Method for Anions in Water by Suppressed Ion Chromatography
D4727 – Standard Specification for Corrugated and Solid Fiberboard Sheet Stock (Container Grade) and Cut Shapes
D5373 – Standard Test Methods for Determination of Carbon, Hydrogen and Nitrogen in Analysis Samples of Coal and Carbon in Analysis Samples of Coal and Coke

(Copies of these documents are available from www.astm.org or ASTM International, 100 Bar Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The material shall be as specified in 3.1.1 through 3.1.2.4.

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3.1.1 Yarn.

3.1.1.1 Type I. Type I packing shall be made of continuous filament graphitic or carbon yarn.

3.1.1.2 Type II. Type II packing shall be made of staple graphitic or carbon yarn. Staple is defined as yarn fabricated using fibers of 2 inch length or longer.

3.1.1.3 Class 1. Class 1 packing shall be made of Type I or II yarn with a pure graphitic dispersion or carbon dispersion, with or without binder or coating (see 3.1.2.3).

3.1.1.4 Class 2. Class 2 packing shall be made of Type I yarn with a pure graphitic dispersion or carbon dispersion, with or without binder or coating (see 3.1.2.3).

3.1.1.5 Class 3. Class 3 packing shall be made of Type I or II yarn with a pure graphitic or carbon dispersion and shall be coated with polytetrafluoroethylene (PTFE), with or without binder (see 3.1.2.3).

3.1.2 Corrosion inhibiting treatments, coatings, binders, and Class 3 coatings.

3.1.2.1 Grade I. Grade I packing shall be provided with a powdered zinc corrosion inhibiting treatment (2 percent Zn, by weight, minimum when tested in accordance with 4.3.11). If no grade is specified, Grade I packing shall be provided. An adhesive meeting the requirements of 3.4.3 may also be used to ensure the adherence of corrosion inhibiting treatments on Grade I packing.

3.1.2.2 Grade N. Grade N packing shall not contain corrosion inhibiting additives.

3.1.2.3 Coatings and binders. Coatings and binders may be utilized to enhance strand integrity. Binders are compounds applied to the strands during braiding. Binding compounds should have no visible thickness and must not obscure the strand intersections (e.g., braiding) or strand surfaces. Coatings are compounds applied to the packing exterior. Coating compounds may obscure the strand intersections and surfaces after braiding. Any coating or binder shall comply with 3.4.3.

3.1.2.4 Class 3 coating. A PTFE coating in accordance with 3.4.2 shall be provided on Class 3 packing only. No other coating or lubricant shall be used.

3.2 Construction. Packing shall be braided for sizes 1/4 inch and over to a diagonal interlocking through-body-to-surface construction. This construction shall use either single or multiple strands on 12, 18, 20, 24 or 36 diagonal and additional center-corner-stuffer strands as necessary to produce a dense square packing with good dimensional stability. Sizes 3/32 through 7/32 inch only may be square braided.

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3.3 Sizes and mass. Packing shall be furnished in the sizes shown in Table I, as specified (see 6.2). Packing shall be formed approximately square in cross section within the dimensional tolerance of Table I. The mass per linear foot shall be not less than that shown in Table I.

3.4 Chemical and physical properties. The properties of the finished packing shall conform to the requirements of Table II. Class 2 material shall also conform to the requirements of Table III.

3.4.1 Dispersion. Dispersion shall be lubrication grade graphite or carbon.

3.4.2 PTFE coating (Class 3 only). The coating shall be from a dispersion of PTFE and shall not exceed 10 percent by mass of the yarn.

3.4.3 Prohibited additions. There shall be no intentional additions of any of the detrimental materials of Table III or antimony, arsenic, bismuth, cadmium, lead, tin or zinc (Grade N only) during the manufacture or packaging of the product.

3.4.4 Braid geometry retention. Untaped, cut ends shall not unravel more than 1/8 inch with the packing dry or wet. Unraveling is a loss of the original braiding shape or dimensions of the cut end extending from the cut for a distance greater than specified above along the axis of the packing.

3.5 Put-up. Unless otherwise specified (see 6.2), packing shall be uniformly coiled on spools or reels in accordance with Table IV. Unless otherwise approved, only one segment of packing shall be provided on each one pound spool for sizes of ½ inch or larger and no more than three segments on each one pound spool for smaller packing sizes.

3.6 Mercury exclusion. During manufacturing, fabrication, handling, packaging, and packing, the packing material shall not come in contact with mercury or mercury containing compounds.

3.7 Surface defects and workmanship. The packing shall be free from extraneous material and visible defects that may affect its serviceability. Defects affecting serviceability include, but are not limited to, severe fraying or fracturing of the filaments, and features such as kinks, bumps or knots which exceed dimensional tolerance limits. The presence of zinc inhibitor on Grade I packing shall be visually apparent and indicated by a dull gray surface color.

3.8 Packing surface tackiness. The top specimen shall lift off cleanly from the bottom specimen with no adhesion when tested in accordance with 4.3.12.

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TABLE I. Sizes and mass of packing.

Normal size (inches)	Tolerance (inches)	Minimum mass (lb/linear ft)
3/32	± 1/64	.0024
1/8	± 1/64	.0045
5/32	± 1/64	.007
3/16	± 1/64	.009
7/32	+ 1/64	.013
1/4	± 1/64	.017
9/32	± 1/32	.021
5/16	± 1/32	.026
11/32	± 1/32	.032
3/8	± 1/32	.038
13/32	± 1/32	.044
7/16	± 1/32	.052
15/32	± 1/32	.059
1/2	± 1/32	.067
17/32	± 1/32	.076
9/16	± 1/32	.085
19/32	± 1/32	.095
5/8	± 1/16	.105
11/16	± 1/16	.127
3/4	± 1/16	.152
13/16	± 1/16	.178
7/8	± 1/16	.206
15/16	± 1/16	.237
1	± 1/16	.270
1 1/4	± 1/16	.421

TABLE II. Chemical and physical properties.

Property	Value	Test
Carbon assay, Type I and II	95 percent min, by mass	4.3.2
Ash	1 percent max, by mass	4.3.3
Specific gravity	1.38 min.	4.3.4
Moisture content	3 percent max	4.3.5
Compression recovery	25 percent min.	4.3.9

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TABLE III. Detrimental materials (Class 2 only).

Element	Maximum allowable impurity levels in parts per million (ppm)
Mercury (Hg)	10
Sulfur (S)	750
Total halogens (chlorine, bromine and fluorine)	500
Chlorine (Cl)	250
Bromine (Br)	250
Fluorine (F)	250

TABLE IV. Put-up

Packing size (inches)	Package (net mass) (lbs)
3/32 through 11/32	1, 2, or 3 lb spools
3/8 through 19/32	2, 3, or 5 lb spools
5/8	3, 5, 10, or 25 lb reels
11/16	3, 5, 10, or 25 lb reels
3/4	5, 10, or 25 lb reels
13/16	5, 10, or 25 lb reels
7/8	5, 10, 25, or 50 lb reels
15/16	5, 10, 25, or 50 lb reels
1	10, 25, or 50 lb reels
1 1/4	10, 25, or 50 lb reels

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements, examinations, and tests as specified herein. Except as otherwise specified in the contract or purchase order, or disapproved by the Government, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements herein. The authorized representative reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Performance of inspection. The performance of the inspections or tests set forth in this specification does not relieve the contractor of his responsibility to provide a product which meets all requirements of this specification. Test specimens shall be stamped or otherwise marked for identification. Any applied marks shall remain on the test specimens until they are

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tested and necessary records made. The test specimens shall not receive any treatment, working, straightening, or other processing that may result in significant change in the properties to be evaluated by the testing, except as provided in specific methods.

4.1.2 Examination on material as offered for acceptance. All applicable examinations and tests required by this specification shall be performed on material as offered for acceptance except testing of material may be performed at an earlier stage of fabrication provided that any subsequent processing does not alter the property or characteristic being tested.

4.1.3 Inspection system requirements. The contractor shall provide and maintain an inspection system acceptable to NAVSEA or its authorized representative for the supplies or services covered by the contract or order. The inspection system shall be in accordance with one of the following, unless an equivalent system is approved:

(1) ISO 9003 (ANSI/ASQC Q9003) with STR-ISO 9000 Supplement C. Supplements B or A are also acceptable.
(2) ISO 9002 (ANSI/ASQC Q9002) with STR-ISO 9000 Supplement C. Supplements B or A are also acceptable.
(3) ISO 9001 (ANSI/ASQC Q9001) with STR-ISO 9000 Supplement C. Supplements B or A are also acceptable.
(4) MIL-I-45208

4.1.3.1 Traceability system. The contractor shall maintain a traceability system to ensure the proper identify of material.

4.1.4 Certification of quality conformance. A certification of quality conformance shall be furnished with each lot of packing offered for acceptance. The certificate shall include identification of any contract assigned part numbers, vendor style or other designations, certification of meeting detrimental material requirements, the quantitative results of specified chemical and physical tests including alternative test methods from those listed in Table VI, and qualitative results of nondestructive tests and examinations. For Grade I packing, the certification of quality conformance shall state that zinc inhibitor has been added in accordance with this specification and no process changes have been made since acceptable zinc inhibitor content testing was performed by the contractor and shall include name of testing laboratory and the acceptable test report number and date.

4.2 Quality conformance inspection. Quality conformance inspection shall be in accordance with Table V and 4.2.2 (see 6.3).

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TABLE V. Quality conformance inspection.

Inspection	Requirements	Tests
Sizes and mass	3.3	4.2.3.3 and 4.3.1
Carbon assay	Table II	4.3.2
Ash	Table II	4.3.3
Specific gravity	Table II	4.3.4
Moisture content	Table II	4.3.5
Detrimental materials (Class 2 only)	Table III	4.3.6
PTFE coating	3.4.2	4.3.7
Surface defects and Workmanship	3.7	4.2.3
Compression recovery	Table II	4.3.9
Braid geometry retention	3.4.4	4.3.10
Zinc content (Grade I only)	3.1.2.1	4.3.11
Packing surface tackiness	3.8	4.3.12

4.2.1 Lot. Unless otherwise specified herein, a lot shall consist of all finished packing of one size, class and grade produced in a continuous run or at the same time under the same conditions. The sampling unit shall be one spool or reel of packing, as applicable.

4.2.2 Sampling. Sampling shall be as specified in 4.2.2.1 and 4.2.2.2.

4.2.2.1 Sampling for visual and dimensional examination. As a minimum, the contractor shall randomly select a quantity of sampling units (spools or reels) from each lot of graphitic or carbon braided yarn packing material in accordance with special inspection level S-4 of ANSI/ASQ Z1.4 and subject them to the examinations of 4.2.3.1, 4.2.3.2, and 4.2.3.3. The contractor has the option of screening 100 percent of the rejected lot for the defective characteristics or providing a new lot, which shall be sampled in accordance with the sampling plan herein.

4.2.2.2 Sampling for quality conformance tests. A single random sample shall be selected from each lot.

4.2.3 Examinations. The units shall be examined for visual and dimensional defects as specified in 4.2.3.1, 4.2.3.2, and 4.2.3.3.

4.2.3.1 Visual examination. Each unit selected in accordance with 4.2.2.1 shall be visually examined to determine conformance to the requirements of 3.7. The presence of corrosion inhibitor on Grade I packing shall be confirmed. Any unit not meeting the requirements of 3.7 shall be rejected, and if the number of defective units in the sample exceeds the acceptance number in ANSI/ASQ Z1.4 for an acceptance quality level (AQL) of 4.0 percent defective, the lot which the sample represents shall be rejected.

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4.2.3.2 Construction. The diagonal interlocking through-body-to-surface or square braided patterns shall be examined visually to determine conformance to 3.2. In addition, a 1 to 2 inch length of sample being examined shall be unbraided and the strands (not yarn plies or ends) counted to determine conformance to 3.2. Any unit not meeting the requirements of 3.2 shall be rejected, and if the number of defective units in the sample exceeds the acceptance number in ANSI/ASQ Z1.4 for an acceptance quality level (AQL) of 4.0 percent defective, the lot which the sample represents shall be rejected.

4.2.3.3 Dimensions. The size of each unit selected for visual examination shall be measured for conformance to the requirements of 3.3 and Table I. A steel slide caliper with 1/64 inch or better divisions shall be used for the dimensional examinations. The size of the unit shall be determined by averaging the measured height and width at the same location. Any unit not meeting the size dimensions specified in Table I shall be rejected, and if the number of defective units in the sample exceeds the acceptance number in ANSI/ASQ Z1.4 for an acceptance quality level (AQL) of 4.0 percent defective, the lot which the sample represents shall be rejected.

4.2.4 Quality conformance tests. The sample selected in accordance with 4.2.2.2 shall be tested as specified in 4.3. Unless otherwise specified herein, all testing shall be conducted on final braided and processed yarn, including any binders, coatings or corrosion inhibitors. If the sample does not conform to all the test requirements, this shall be cause for rejection of the lot which it represents.

4.3 Tests. Tests shall be conducted as specified in 4.3.1 through 4.3.12.

4.3.1 Mass. The mass shall be determined using a specimen at least 2 feet in length for measurement in conjunction with the values of Table I. The specimen shall be measured using a steel rule with 1/8 inch divisions accurate to 1/8 inch. The mass shall be determined to the nearest 0.005 pound (see 3.3).

4.3.2 Carbon assay. The sample shall be dried to a constant mass at 149 ± 3°C (300 ± 5°F) before testing. For Grade I packing only, the test may be conducted prior to the corrosion inhibiting treatment or the added mass of the corrosion inhibitor may be determined and subtracted from the base mass of the sample. The percent carbon shall be based upon mass of the dried sample. This determination shall be made in accordance with ASTM D5373 or an alternate method of analysis with equal or improved accuracy and precision (see Table II).

4.3.3 Ash content. The ash content shall be determined in accordance with ASTM C561 (see Table II). The ash content of Grade I packing may be determined after all processing but prior to the application of the zinc inhibitor.

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4.3.4 Specific gravity. The specific gravity shall be determined in accordance with ASTM C135, modified as follows (see Table II):

A 50 gram sample shall be prepared for grinding by unbraiding, cutting, slicing, or otherwise reducing the braided packing to pieces not larger than 1/8 inch.

4.3.5 Moisture content. The moisture content shall be determined in accordance with ASTM C562 (see Table II).

4.3.6 Detrimental material tests. For determination of the detrimental materials listed in Table III for Class 2 only, the test methods of Table VI or alternate methods of equal or improved accuracy and precision, such as other applicable ASTM methods, shall be used). When an alternate method is used, it shall be identified on the certificate of quality conformance.

Table VI. Detrimental material tests.

Element	Preparation/Analysis Test Methods
Chlorine (Cl))	(1) ASTM C1502 (sample preparation and analysis) (2) ASTM D129 / ASTM D512 (3) ASTM D808/ASTM D4327 (ion chromatography)
Bromine (Br)	(1) ASTM D129 / ASTM D512 (2) ASTM D808/ASTM D4327 (ion chromatography)
Fluorine (F)	(1) ASTM C1502 (sample preparation and analysis (2) ASTM D129 / ASTM D1179 (3) ASTM D3761 (sample preparation and analysis) (4) ASTM D808/ASTM D4327 (ion chromatography)
Sulfur (S)	(1) High temperature combustion in 100% Oxygen/Non-Dispersive Infrared Analysis or Ion chromatographic Analysis (2) ASTM C816 (sample preparation and analysis) (3) ASTM D4239 Method #A (sample preparation and analysis) (4) ASTM D808/ASTM D516 (turbidemetric), ASTM D4327 (ion chromatography)
Mercury (Hg)	(1) Direct analysis of volatile elements (Hg) by Emission Spectrographic Method (2) Direct analysis of volatile elements by atomic absorption per ASTM D3684.
NOTE: Where two preparation or analysis test methods are specified, either one is acceptable.	

The vendor shall provide certification that the limits of Table III have been met and that low melting metals (Sb, As, Bi, Cd, Pb, Sn and Zn [Zn for Grade N only]) have not been added as intentional constituents. In lieu of specific test results for the lot to be delivered, certifications

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may be based on tests of both raw materials and production lots of similar finished packing material over an extended time period (not exceeding three years between tests). Any change to manufacturing processing which affects product composition, including changes to raw material, binders or inhibitors (Grade I) shall require additional testing to form the basis for future certifications.

4.3.7 Analysis for PTFE coating. A 5 gram specimen of packing shall be placed in a crucible and heated at 104 ± 1°C (220 ± 2°F) to constant mass (original dry mass) at room temperature. Then the specimen shall be heated at 316 ± 5.5°C (600 ± 10°F) for 24 hours, cooled, and the mass determined. The same specimen shall then be heated at 482 + 14°C (900 + 25°F) for 3 hours, cooled, and the mass determined. After the 482°C (900°F) heating, the crucible mass shall be redetermined without the specimen unless a platinum crucible was used. The new mass of the crucible shall be used to determine the mass of the residue after heating. Heating shall be done in a ventilated hood to avoid exposure to toxic vapors. The percentage of PTFE shall be calculated as follows, based on an average of three determinations (see 3.4.2).

$$\text{Percent PTFE} = \frac{100\,(F-N)}{W}$$

Where:

W = Average original dry mass of specimens after extraction
F = Average mass of residue in crucibles after heating at 316°C (600°F)
N = Average mass of residue in crucibles after heating at 482°C (900°F)

4.3.8 Staple fiber lengths. Staple fiber length shall be determined in accordance with ASTM D1234 (see 3.1.1).

4.3.9 Compression recovery. A compression recovery test procedure shall be prepared and shall include set-up, equipment, fixturing, and test process steps as a minimum. The procedure shall be made available for review upon request. Three specimens, each 1.75 ± 0.125 inch long, shall be prepared from the sample and the results of the following test of each specimen averaged for comparison with the compression recovery limit of Table II. Each specimen shall be visually examined after test and the lot shall be rejected if any of the specimens have split or unraveled more than three times the nominal size or one half the length of a braided yarn specimen, whichever is less.

4.3.9.1 Test procedure. Center the specimen on a flat plate which is larger than the specimen and has its upper surface perpendicular (± 2°) to the load to be applied. Place a flat plate of similar size on top of the specimen so that its lower surface is parallel (± 2°) to the upper surface of the bottom plate. The upper plate shall weigh 5 ± 1 pounds. Both plates shall be rigid and maintained parallel during the testing. After the weight of the upper plate has been applied to the specimen for at least 15 seconds, measure and record the thickness (P) of the preloaded specimen. Several measurements about the circumference of the test plate should be taken and averaged for this, and subsequent, measurements.

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Apply the major load, by either the top or bottom plate, along an axis passing through the center of the specimen in a slow, uniform manner so that the major load is attained within 20 ± 5 seconds. The major load is that load which will produce a stress of 4500 pounds per square inch ± 10 percent on the area of the specimen initially in contact with the lower test plate.

Maintain the major load for at least 60 seconds and measure and record the thickness (M) of the fully loaded specimen.

Immediately remove the major load and after about 60 seconds measure and record the recovered thickness (R) under the original preload.

Calculate the percent compression recovery (PCR) as follows:

$$PCR = ((R - M)/(P - M)) * 100$$

4.3.10 Braid geometry retention. Cut a 2 ± 1/16 inch length of packing. (The ends shall not be taped prior to or after cutting). Soak the test piece in room temperature tap water for 15 minutes, minimum. The yarn shall not unravel at the ends more than a total of 0.25 inches (0.125 inches at either end).

4.3.11 Zinc content. The zinc content shall be determined by the Inductively Coupled Plasma Atomic Emission Spectroscopy (ICP) method. An alternative equivalent test method is allowed when approved in writing prior to use. After zinc analysis is acceptably performed once for each size, provided the internal procedure relative to zinc processing has not been changed, a certification of quality conformance may be accepted thereafter. A process change constitutes a different grade and particle size of powder, different binder, or different application method.

4.3.11.1 Test procedure. To minimize variations between test facilities, the following shall be observed when using the ICP method:

(a) Special care shall be taken not to cause handling loss of the zinc inhibitor from the packing.
(b) Grind/crush/cut-up the complete sample. Minimize contamination of the sample, such as avoiding the use of zinc plated instruments.
(c) The sample shall be subjected to standard laboratory digestion procedures using nitric acid prior to analysis by ICP.

4.3.12 Packing surface tackiness. The following testing is to be performed at 23 °C +/- 5 °C (73 °F +/- 9 °F) and less than 70% relative humidity, and for Grade I prior to application of the corrosion inhibitor or adhesive. Cotton gloves must be worn when handing the sample prior to removal of the test specimens and during testing. Testing is to be performed on a scale to measure the load required to achieve the below specified contact pressure.

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(a) Remove two specimens from the sample, each the same length between 1 and 2 inches long.
(b) Please a piece of paper on the scale to prevent specimen sticking, then place one of the specimens on the paper. This is the bottom specimen.
(c) Place the second (top) specimen on top and parallel to the bottom specimen so they fully overlap.
(d) Using gloved hands to prevent sticking, press the top specimen on to the bottom specimen with sufficient load to generate a minimum of 40 psi contact pressure between the specimens (load divided by (specimen length X nominal size)), and hold for a minimum of 10 seconds. Fixturing may be used to stabilize the specimens so long as there is no interference with load application.
(e) Slowly lift up the gloved hands to remove the load leaving the top specimen sitting on the bottom specimen. Grasp only the top specimen and lift up. Check for adhesion to the bottom specimen to the acceptance criteria in 3.8.
(f) The test is invalid and must be repeated with new specimens if the top specimen comes off the bottom specimen when the load is removed.

4.4 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of Section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 General.

5.1.1 Navy Fire-retardant requirements.

(a) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B – Type II - weather resistant. Category 1 - general use.
Level C – Type I - non-weather resistant. Category 1 - general use.

(b) Fiberboard. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements of ASTM D4727.

5.2 Preservation. Preservation shall be level A, C, or commercial as specified (see 6.2).

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5.2.1 Level A. Packing, put-up as specified in 3.5, shall be protected by placing a wrap of greaseproof barrier material over the packing material and between the flanges of the spool or reel. The wrap shall be secured, utilizing pressure-sensitive tape. Packing material furnished in put-up form other than on spools or reels shall be completely wrapped in barrier material and secured with tape as specified herein. The greaseproof barrier herein is not required when the unit container selected (see 5.2.1.1) is treated for greaseproofness or is inherently greaseproof. To avoid shipping and handling loss of zinc inhibitor, shrink wrapping of Grade I packing is prohibited.

5.2.1.1 Unit pack. Unit packs (unit containers) shall conform to MIL-STD-2073-1; Appendix F, Table I. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option. Containers shall be of type weather resistant and conform to closure method V in accordance with the appendix of the box specification.

5.2.2 Level C. Graphitic and carbon braided yarn shall be preserved as specified for level A except that the unit containers specified in 5.2.1.1 shall be of the non-weather resistant type, class, or variety as applicable. Fiberboard box closure shall be in accordance with method I of the box specification, using pressure sensitive tape.

5.2.3 Commercial. Commercial preservation shall be in accordance with ASTM D3951.

5.3 Packing. Packing shall be level A, B, C, or commercial, as specified (see 6.2).

5.3.1 General. Shipping containers shall contain identical quantities of identical material and shall be of minimum weight and cube, similar construction, and of uniform size.

5.3.2 Levels A, B, and C containers. Material preserved as specified (see 5.2) shall be packed in shipping containers for the level of packing specified (see 6.2), in accordance with MIL-STD-2073-1; Appendix C, Table VII. Unless otherwise specified (see 6.2), container selection shall be at the contractor's option.

5.3.2.1 Caseliners, closure and gross weight. Caseliner, closure, and gross weight shall be as specified in 5.3.2.1.1, 5.3.2.1.2, 5.3.2.1.3.

5.3.2.1.1 Caseliners. Unless otherwise specified (see 6.2), level A shipping containers containing level C or commercial unit pack containers shall be provided with waterproof caseliners in accordance with MIL-STD-2073-1.

5.3.2.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that class weather resistant fiberboard boxes shall be closed in accordance with method V and reinforced with nonmetallic or tape banding, and non-weather resistant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

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5.3.2.1.3 Weight. Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.3 Commercial. Material preserved as specified (see 6.2) shall be packed for shipment in accordance with ASTM D3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with the minimum of 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base that will support the material and facilitate handling by mechanical handling equipment during shipment.

5.4 Marking. Marking shall be as specified in 5.4.1.

5.4.1 Levels A, B, C, and commercial. Item description marking shall include, as a minimum, the size, type, class and grade of the graphitic or carbon braided yarn packing. Material provided to previous revisions, and not marked to indicate the grade classification added by this revision, shall be considered to be Grade N. In addition to any special marking required (see 6.2), level A, B, and C interior packs and shipping containers shall be marked in accordance with MIL-STD-2073-1, Appendix F, and commercial interior packs and shipping containers shall be marked in accordance with ASTM D3951. In addition, bar coding shall be applied in accordance with the marking requirements of MIL-STD-2073-1. In addition to marking on the packaging and packing (container), individual spools shall be marked, as a minimum, with the packing size, class, grade, Commercial and Government Entity (CAGE) Code of manufacturer and, if other than manufacturer, the packaging activity, National Stock Number, and contract number.

6. NOTES

6.1 Intended use. Type I and Type II packings are intended for general shipboard service as anti-extrusion rings used in conjunction with corrugated ribbon or preformed ring graphitic packing material in valves. Grade I packing is intended for use in valves with non-corrosion resistant (for example, carbon steel, 400 series stainless steel) stem and packing gland parts.

6.1.1 Class 1. Class 1 packing is intended for use in superheated steam, feed-water, and condensate systems.

6.1.2 Class 2. Class 2 packing is intended for use in saturated steam, feed-water, and condensate systems in special applications.

6.1.3 Class 3. Class 3 packing is intended for use in steam, feed-water, and condensate systems with stuffing box temperatures limited to 260°C (500°F).

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6.2 Ordering data. Procurement documents should specify the items below. Ordering data options identified by a "+" may not be specified by NAVSEA or its authorized representative. Accordingly, the purchaser should ensure that the items identified by a "+" are specified in any subtier procurements.

+ (a) Title, number, and date of this specification.
(b) Type, class and grade required (see 1.2).
(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
(d) Size required (see 3.3).
(e) Put-up, if other than required by Table IV (see 3.5).
(f) When fire-retardant materials are not required (see 5.1.1).
(g) Level of preservation and level of packing required (see 5.2, 5.2.1, 5.2.2, 5.2.3, 5.3, 5.3.2 and 5.3.3).
(h) Container selection, if other than contractor's option (see 5.2.1.1 and 5.3.2).
(i) When caseliners are not required (see 5.3.2.1.1).
(j) Special markings required (see 5.4.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.2	DI-RELI-80939	Test and inspection report	-----

The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Subject term (key word) listing.

Carbon assay
Center-corner-stuffer strands
Mass
PTFE coating
Reflux condenser

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5330-2019-024)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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